

# Worksheet — Pulleys and Hanging Masses Simulation

Use with *Simulation - Pulleys and hanging masses.html* · one pulley (Atwood machine) and two or three pulleys with a knot · Newton's laws (Blocks 3–4)

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Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

## How to use this worksheet

Scan the QR code or open [biba0112.github.io/physics-sims/23-pulleys-hanging-masses/](https://biba0112.github.io/physics-sims/23-pulleys-hanging-masses/) in a browser (offline: *Simulation - Pulleys and hanging masses.html*). Choose the **number of pulleys** at the top: ① one pulley (Atwood machine) · ② two pulleys and a knot · ③ three pulleys and a knot. Pick an **Example** or set the masses yourself (type a value in the box next to a slider).



Scan to open

① Press **Release**; ◀/▶ step through the motion. ② ③ **Drag a pulley** or change a mass: the knot swings and settles. Read the angles on the protractor around the knot.

**Part 1** before the lab ( $\approx 15$  min) · **Part 2** during the lab ( $\approx 25$  min) · **Part 3** homework ( $\approx 30$  min).  $g = 9.81 \text{ N kg}^{-1}$ . Answers to 3 s.f.

## Part 1 — Pre-lab: predict, then test

| Question                                                                                                                                                               | My prediction (before) | What the simulation showed | Physics explanation |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|---------------------|
| 1. ① 0.200 kg and 0.220 kg hang over one pulley. While they move, is the tension equal to the lighter weight, between the two weights, or equal to the heavier weight? |                        |                            |                     |
| 2. ① Two equal masses hang over the pulley. You push one down and let go. Does it slow down, keep a constant speed or speed up?                                        |                        |                            |                     |
| 3. ② 0.30 kg, 0.40 kg and 0.50 kg (the 0.50 kg hangs from the knot). Is the angle between strings 1 and 2 less than, equal to or more than $90^\circ$ ?                |                        |                            |                     |
| 4. ② Now $m_3 = 0.80 \text{ kg}$ ( $m_1 = 0.30 \text{ kg}$ , $m_2 = 0.40 \text{ kg}$ ). What happens to the knot?                                                      |                        |                            |                     |

**5. (Card C2★)** Masses of 2.0 kg and 3.0 kg hang over a light, smooth pulley. Calculate (a) the acceleration and (b) the tension. Check: ① → Example Card C2★.

Working: FBD of each mass →  $\Sigma F = ma$  for each (or the whole system)

**6. (3-4-5 knot)** Use question 3: the strings meet at  $90^\circ$ . Calculate the angle between the string of the 0.30 kg mass and the vertical. Check: ② → Example 3-4-5; read  $\theta_1$ .

Working: FBD of the knot →  $\Sigma F_x = 0$

## Part 2 — During the lab

### A • Atwood machine: measure $g$

Pulley on a clamp, two hangers with slotted masses, drop height  $h = 1.00$  m (metre rule), stopwatch or light gate. Keep  $m_1 + m_2 = 0.420$  kg: move 5 g or 10 g slotted masses from one side to the other. Time the drop three times and take the mean.

**7.** For 0.200 kg + 0.220 kg, calculate the acceleration a light, smooth pulley would give and the time to fall 1.00 m.

Working

| run | $m_1 / \text{kg}$ | $m_2 / \text{kg}$ | $x = (m_2 - m_1)/(m_1 + m_2)$ | $t_1 / \text{s}$ | $t_2 / \text{s}$ | $t_3 / \text{s}$ | mean $t / \text{s}$ | $a = 2h/t^2 / \text{m s}^{-2}$ | sim $a / \text{m s}^{-2}$ |
|-----|-------------------|-------------------|-------------------------------|------------------|------------------|------------------|---------------------|--------------------------------|---------------------------|
| 1   |                   |                   |                               |                  |                  |                  |                     |                                |                           |
| 2   |                   |                   |                               |                  |                  |                  |                     |                                |                           |
| 3   |                   |                   |                               |                  |                  |                  |                     |                                |                           |
| 4   |                   |                   |                               |                  |                  |                  |                     |                                |                           |
| 5   |                   |                   |                               |                  |                  |                  |                     |                                |                           |

8. Plot  $a$  (y-axis) against  $x$  on graph paper and draw the best-fit line. Gradient = \_\_\_\_\_  $\text{m s}^{-2}$  intercept = \_\_\_\_\_  $\text{m s}^{-2}$

Compare your gradient with  $9.81 \text{ m s}^{-2}$ : percentage difference = \_\_\_\_\_. In the simulation: ● Record five runs, Data tab →  $a-x$ .

9. **Predict, then test.** A real pulley has friction (set *Pulley friction* = 0.02 N and record the five runs again). The  $a-x$  line: ☐ is steeper ☐ has the same gradient but moves down ☐ goes through the origin with a smaller gradient. Explain:

Explanation

## B · Pulley board: three forces in balance

Two pulleys clamped to a vertical board, three hangers tied to one knot, a protractor behind the knot. Let the knot settle, tap the board gently, then read each string's angle to the vertical (plumb line).

| set-up | $m_1 / \text{kg}$ | $m_2 / \text{kg}$ | $m_3 / \text{kg}$ | $\theta_1$ measured | $\theta_2$ measured | $\theta_1$ sim | $\theta_2$ sim | angle 1-2 |
|--------|-------------------|-------------------|-------------------|---------------------|---------------------|----------------|----------------|-----------|
| 3-4-5  | 0.30              | 0.40              | 0.50              |                     |                     |                |                |           |
| equal  | 0.30              | 0.30              | 0.30              |                     |                     |                |                |           |
| own    |                   |                   |                   |                     |                     |                |                |           |
| own    |                   |                   |                   |                     |                     |                |                |           |

10. For each row, calculate the horizontal components  $T_1 \sin \theta_1$  and  $T_2 \sin \theta_2$  ( $T = mg$ ) and the vertical total  $T_1 \cos \theta_1 + T_2 \cos \theta_2$ . Do they show  $\Sigma F_x = 0$  and  $\Sigma F_y = 0$  within your reading uncertainty ( $\pm 1^\circ$ )?

Working

**11. Predict, then test.** You move one pulley higher on the board. When the knot settles again, are the angles between the strings different, the same, or all  $120^\circ$ ? Drag a pulley in the simulation to check, then explain.

Prediction and explanation

### Part 3 — Homework: analyse

**12. (Problem set Q27)** With  $m_1 = 0.190 \text{ kg}$  and  $m_2 = 0.210 \text{ kg}$  a student measures  $a = 0.48 \text{ m s}^{-2}$ . (a) Determine  $g$  from this run. (b) Assume the difference from  $9.81 \text{ m s}^{-2}$  is caused by friction in the pulley. Calculate the friction force. Check: ① → *Problem set Q27*.

Working

**13. (Card C2★★★★)** Masses of  $1.20 \text{ kg}$  and  $0.80 \text{ kg}$  hang over a smooth pulley; the  $1.20 \text{ kg}$  mass starts  $0.60 \text{ m}$  above the floor. Calculate its speed when it hits the floor. Check: ① → *Card C2★★★★*, step to the end.

Working

**14. (FBD guide)** Masses of  $1.0 \text{ kg}$  and  $2.0 \text{ kg}$  hang over a smooth pulley. Calculate the tension and show that it lies between the two weights.

Working

**15.** ②  $m_1 = 0.30 \text{ kg}$ ,  $m_2 = 0.40 \text{ kg}$  and  $m_3 = 0.60 \text{ kg}$ . Sketch the vector triangle and calculate the angle between strings 1 and 2. Check: ②  $\rightarrow 0.30, 0.40, 0.60 \text{ kg}$ .

Working

**16. (Problem set Q19)** A 15 kg shop sign hangs from two cables, each at  $40^\circ$  to the horizontal. Draw the FBD of the knot and calculate the tension in each cable. Check: ②  $\rightarrow$  Problem set Q19 (the two 11.67 kg masses play the cables).

Working

**17.** ③ Three equal masses hang over three pulleys. Explain, using a vector triangle, why the angle between every pair of strings is  $120^\circ$ , wherever the pulleys are (as long as the knot can balance).

Working